Heritage Bank — Domain Context

About the Organisation

Heritage Bank & Trust is a mid-sized regional bank with **$48 billion in assets**, operating **210 branches** across the Midwest and Southeast US. Founded in 1923, Heritage runs its core banking operations on **IBM AS/400 (IBM i)** — a platform that has been in continuous production for 31 years. The bank serves approximately **1.2 million retail customers** and **85,000 business banking clients**.

Heritage Digital is the internal programme to modernise customer-facing channels — mobile banking, online banking, and a new SMB business portal — while leaving the proven AS/400 core banking system intact and treating it as the authoritative source of record.

The programme is a **strangler-fig modernisation**: new capabilities are built on a modern microservices layer that integrates to AS/400 via a **Banking Integration Hub (BIH)**, progressively replacing screen-scraping legacy batch integrations one domain at a time.

Business Domain Concepts

User Personas

Persona	Description	Key Needs
Retail Customer	Individual account holder	Balance checks, transfers, bill pay, statements, card controls
SMB Owner	Small-medium business banking client	Multi-account view, ACH payroll, wire transfers, account user management
SMB Authorised User	Employee granted access by SMB Owner	Role-scoped actions (view-only, initiate-only, approve-only)
Branch Teller	In-branch staff handling transactions	Look up customer, initiate transactions on behalf of customer, override certain holds
Branch Manager	Supervises tellers, handles exceptions	Override approvals, large cash transactions, account opening authority
Customer Service Agent (CSA)	Phone/chat support	Read-only customer view, initiate refunds, raise service tickets
Fraud Analyst	Monitors transaction patterns	Alert queue, case management, card block/unblock
Compliance Officer	AML, BSA, regulatory reporting	SAR filing, CTR generation, audit trail access

Account Types

- **Checking** — Demand deposit account (DDA); standard and interest-bearing variants
- **Savings** — Regulation D compliant; 6 withdrawal limit per statement cycle (legacy rule still enforced)
- **Money Market** — Tiered interest, cheque-writing capability

- **Certificate of Deposit (CD)** — Fixed term (3, 6, 12, 24, 36 months); early withdrawal penalty
- **HELOC** — Home Equity Line of Credit; draw/repayment cycle; linked to real estate collateral in AS/400
- **Personal Loan** — Instalment loan; amortisation schedule from AS/400 Loan module
- **Business Checking / Business Savings** — Multi-signatory; ACH origination enabled
- **Escrow / Trust accounts** — Managed by Wealth division (separate system, out of scope Phase 1)

Core Banking on IBM AS/400 (IBM i)

- Core system: **FiServ Precision** running on IBM i 7.4 (Power9 hardware)
- All account records, transaction ledger, interest calculation, and GL entries live in AS/400
- Batch processing windows:
  - **Nightly batch**: midnight–4 AM CT — ACH settlement, interest posting, statement generation
  - **End-of-day (EOD)**: 6 PM CT — balance cutoff for same-day transactions
  - **Intra-day**: ACH returns processed at 10 AM and 2 PM CT
- Interfaces into AS/400:
  - **Screen scraping (legacy)**: IBM i green-screen sessions via TN5250 terminal emulation — used by older internal tools
  - **FiServ Open Solutions API**: REST wrapper around FiServ Precision; covers ~60% of operations; remainder requires COBOL batch jobs
  - **MQ Series**: IBM MQ for high-reliability async message exchange between AS/400 and modern services
- Real-time balance: AS/400 provides **book balance** and **available balance** (after holds); ledger balance updated at EOD batch

Payments & Transfers

- **Internal transfers**: real-time within Heritage accounts; posted immediately to AS/400 via FiServ API
- **ACH (Automated Clearing House)**: originated via FedACH; files built by Payment Service, submitted via AS/400 ACH module; next-day or same-day (SDIE) settlement
- **Wire transfers**: domestic (Fedwire) and international (SWIFT); manual approval for wires > $10,000; processed by Wire Desk
- **Bill Pay**: managed by **Fiserv CheckFree**; Heritage sends payee data, CheckFree handles disbursement
- **Zelle**: peer-to-peer via Early Warning Services network; real-time; integrated at API layer (not through AS/400)
- **RTP (Real-Time Payments)**: TCH RTP network; sub-5-second settlement; routing via Payment Service
- **Card payments**: debit card managed by **FIS Worldpay**; Heritage AS/400 holds card-to-account linkage; Worldpay handles authorisation and settlement

Lending

- **Consumer loan origination**: Blend platform (cloud-native); underwriting decision from Experian + Equifax; approval posts loan to AS/400 Loan module
- **Mortgage**: Encompass LOS; out of scope for Digital programme (separate team)
- **SMB lending**: manual process in branches; digital application form feeds AS/400 via BIH

Fraud & Risk

- **Transaction monitoring**: NICE Actimize running on-prem; receives real-time transaction feed from Payment Service via Kafka

- **Card fraud**: FIS Worldpay fraud scoring; Heritage Fraud Analyst can block/unblock cards via Card Management Service
- **Suspicious Activity Reports (SARs)**: filed via FinCEN BSA E-Filing; triggered by Actimize case + Compliance Officer approval
- **Currency Transaction Reports (CTRs)**: auto-generated for cash transactions ≥ $10,000 in a single day
- **Velocity rules**: configurable per product (e.g., Zelle: max $2,500/day; ACH credit: max $25,000/day)

Digital Channels

- **Mobile App**: iOS and Android (React Native); biometric auth; current DAU ~180,000
- **Online Banking**: React web SPA; session-based auth with Okta
- **Branch Teller System**: internal web app; read/write via BIH; replaces green-screen for most workflows
- **IVR**: Nuance-hosted phone banking; balance and recent transactions via AS/400 direct integration (not being replaced)

Compliance & Regulatory Context

Bank Secrecy Act (BSA) / AML

- Customer Identification Programme (CIP): identity verified at account opening; documents stored in customer record
- Ongoing monitoring: NICE Actimize monitors all transactions for suspicious patterns
- SAR filing: required within 30 days of determination; 90-day extension available with documentation
- CTR filing: required within 15 days of qualifying cash transaction
- Record retention: 5 years for BSA records

Regulation E (Electronic Fund Transfers)

- Unauthorised transaction disputes: provisional credit within 10 business days; investigation within 45 days
- Error resolution notices: written response to customer within 3 business days of investigation completion

Regulation D (Savings Accounts)

- Maximum 6 "convenient" withdrawals per statement cycle from savings/money market
- Excess transactions: fee on 7th+ transaction; account may be converted to checking after repeated violations

Dodd-Frank / Consumer Protection

- Overdraft opt-in required for ATM and one-time debit card transactions
- Reg CC: funds availability schedule; next-day availability for government cheques, $225 same-day rule

SOX (Sarbanes-Oxley)

- IT General Controls (ITGC) audit annually; change management evidence required for all production changes
- Segregation of duties: no developer has production AS/400 access; change requires CAB approval
- Audit trail: all configuration changes to BIH and modern services logged with actor, timestamp, diff

PCI DSS

- Card data handled by FIS Worldpay; Heritage is SAQ-D (card data in scope via network)
- Tokenisation: PAN replaced with Worldpay token at point of capture; token stored in Card Management Service
- Quarterly ASV scans; annual ROC assessment

Accessibility

- ADA / Section 508 for digital channels: WCAG 2.1 AA
- Mobile app: VoiceOver and TalkBack support mandatory

Business Rules

1. Transfers between Heritage accounts posted immediately; available balance updated in real time via FiServ API — **book balance updated at EOD batch**
2. Wire transfers above **$10,000** require dual authorisation: initiating CSA + Wire Desk supervisor
3. ACH same-day credit (SDIE) cut-off is **2:30 PM CT**; submissions after cut-off are queued for next business day
4. Overdraft protection: if linked savings has sufficient funds, transfer occurs automatically (max **3 transfers per day**); otherwise standard NSF fee of $35 applies
5. CD early withdrawal penalty: **90 days' interest** for terms ≤ 12 months; **180 days' interest** for terms > 12 months
6. Zelle payments to new recipients (first payment) are **held for 15 minutes** for fraud review before release
7. SMB users can be granted one of three roles: **Viewer** (read-only), **Initiator** (can create but not approve), **Approver** (can approve up to their limit); no user can be both Initiator and Approver on the same transaction
8. Failed logins: **5 consecutive failures** lock the account; mobile biometric bypass disabled after lockout; unlock via verified phone call or branch visit
9. Savings account excess withdrawal: **fee of $10** applied per transaction over 6; if limit exceeded in **3 consecutive cycles**, account automatically converted to checking with written notice

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Integration Landscape

System	Purpose	Protocol
IBM AS/400 (IBM i) — FiServ Precision	Core banking: accounts, ledger, interest, GL	FiServ Open Solutions API (REST) + IBM MQ
Banking Integration Hub (BIH)	API gateway / adaptor layer between AS/400 and modern services	REST / Kafka / IBM MQ
FIS Worldpay	Debit card issuer processing, authorisation, settlement	REST API + ISO 8583
Fiserv CheckFree	Bill pay disbursement	Proprietary file exchange (SFTP)
Early Warning Services (EWS)	Zelle P2P network	REST API

TCH RTP	Real-Time Payments network	ISO 20022 / REST
NICE Actimize	AML / transaction monitoring	Kafka (real-time feed) + REST (case management)
Blend	Consumer loan origination	REST API
Experian / Equifax	Credit bureau decisioning	REST API
Okta	Identity provider, SSO, MFA	OIDC / SAML 2.0
Nuance (IVR)	Phone banking (not being replaced)	Direct AS/400 CICS integration
Salesforce Financial Services Cloud	CRM, service cases, branch activity log	REST API (Salesforce Connect)
Snowflake	Data warehouse for analytics and regulatory reporting	Kafka → Snowflake Connector